

SERGEY PROKUDIN

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Universitätstrasse 6, Zürich, Switzerland, 8092

Homepage Google Scholar GitHub

APPOINTMENTS

Eidgenössische Technische Hochschule (ETH), Zürich

Senior Scientist and Lecturer

Postdoctoral Researcher

Computer Vision and Learning Group (Prof. Dr. Siyu Tang)

March 2024 – Present

January 2021 – March 2024

EDUCATION

Max Planck Institute for Intelligent Systems, Tübingen

PhD, thesis: “Robust and Efficient Deep Visual Learning”

Perceiving Systems Department (Prof. Dr. Michael J. Black)

Supervisors: Dr. Peter Gehler, Dr. Sebastian Nowozin

January 2016 – December 2020

Lomonosov Moscow State University

Diploma of Higher Education (MSc equivalent)

Faculty of Computational Mathematics and Cybernetics

September 2005 – June 2010

RESEARCH STATEMENT

I work across computer vision, computer graphics, and machine learning. My research focuses on representations of real-world 3D and 4D phenomena: reconstructing geometry, modelling motion, rendering photorealistic scenes, and extracting structure from visual observations.

I am increasingly focused on geometry-grounded reasoning: the bridge from 3D and 4D reconstruction to higher-level inference. My goal is to build models that form and update internal maps to answer three questions: what is where, what changes, and how a machine can interact with its environment.

SELECTED PUBLICATIONS

Full list on Google Scholar.

[1] J. Gajardo, M. Volpi, M. Mihajlovic, Siyu Tang, L. Roth, and **Sergey Prokudin**. “GrowFields: Compositional 4D Neural Fields for Topology-Changing Plant Growth.” *European Conference on Computer Vision (ECCV)*, 2026.

[2] Y. Chen, Y. Wang, X. Zhang, **Sergey Prokudin**, and Siyu Tang. “GGPT: Geometry-Grounded Point Transformer.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

[3] F. Rajič, H. Xu, M. Mihajlovic, S. Li, I. Demir, E. Gündoğdu, L. Ke, **Sergey Prokudin**, Marc Pollefeys, and Siyu Tang. “Multi-view 3D Point Tracking.” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. **(Oral)**

[4] Y. Chen, M. Mihajlovic, X. Chen, Y. Wang, **Sergey Prokudin**, and Siyu Tang. “SplatFormer: Point Transformer for Robust 3D Gaussian Splatting.” *International Conference on Learning Representations (ICLR)*, 2025. **(Spotlight)**

- [5] M. Mihajlovic, **Sergey Prokudin**, Siyu Tang, R. Maier, F. Bogo, T. Tung, and E. Boyer. “Splat-Fields: Neural Gaussian Splats for Sparse 3D and 4D Reconstruction.” *European Conference on Computer Vision (ECCV)*, 2024.
- [6] X. Chen, M. Mihajlovic, S. Wang, **Sergey Prokudin**, and Siyu Tang. “Morphable Diffusion: 3D-Consistent Diffusion for Single-image Avatar Creation.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [7] M. Mihajlovic, **Sergey Prokudin**, Marc Pollefeys, and Siyu Tang. “ResFields: Residual Neural Fields for Spatiotemporal Signals.” *International Conference on Learning Representations (ICLR)*, 2024. **(Spotlight)**
- [8] **Sergey Prokudin**, Qianli Ma, Maxime Raafat, Julien Valentin, and Siyu Tang. “Dynamic Point Fields.” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. **(Oral)**
- [9] K. Karunratanakul, **Sergey Prokudin**, Otmar Hilliges, and Siyu Tang. “HARP: Personalized Hand Reconstruction from a Monocular RGB Video.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [10] **Sergey Prokudin**, Michael J. Black, and Javier Romero. “SMPLpix: Neural Avatars from 3D Human Models.” *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2021.
- [11] **Sergey Prokudin**, Christoph Lassner, and Javier Romero. “Efficient Learning on Point Clouds with Basis Point Sets.” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- [12] **Sergey Prokudin**, Peter Gehler, and Sebastian Nowozin. “Deep Directional Statistics: Pose Estimation with Uncertainty Quantification.” *European Conference on Computer Vision (ECCV)*, 2018.

SUPERVISION & LEADERSHIP

Project Leader: “Surgical Digital Twins” subproject of PROFICIENCY, an Innosuisse Flagship Project on data-driven surgical training. Led the research concept for the SurgiScope 3D training platform, developed with ZHAW and Balgrist.

Student Supervision: co-supervised PhD students and 20+ MSc, BSc, and semester projects at ETH Zürich, several resulting in publications: Marko Mihajlovic [5, 7], Yutong Chen [2, 4], Xiyi Chen [6], Frano Rajič [3], and Joaquin Gajardo [1].

TEACHING EXPERIENCE

Lecturer:

“Computer Vision”

Autumn Semester 2024, 2025

“Digital Humans”

Spring Semester 2024, 2025, 2026

Lead Teaching Assistant:

“Digital Humans”

Spring Semester 2023

Teaching Assistant:

“Linear Algebra”

Autumn Semester 2021 – 2023

INVITED TALKS

Applied Machine Learning Days 2026, Lausanne: “Foundation Models for Surgical Digital Twins” (Swiss AI Initiative track).

OTDigital 2026, Berlin: “Photorealistic Digital Twins for Orthopedic Surgical Training.”

Informatiktage 2025, Zürich: “Revolutionizing Surgical Education with Digital Twins.”

Medical Augmented Reality Summer School 2023: “Neural Radiance Fields.”

CVPR 2021 Tutorial (“SMPL made simple”): “Combining SMPL and Neural Rendering.”

INDUSTRY EXPERIENCE

Amazon, Body Labs *July 2019 – December 2019*
Applied Science Intern: neural human avatar rendering (publication [10]).

Amazon, Body Labs *August 2018 – January 2019*
Applied Science Intern: point-cloud encoding for 3D shape analysis (publication [11]).

Kaspersky Lab, Detection Methods Analysis Team *April 2013 – December 2015*
Senior Research Developer *April 2011 – March 2013*
Research Developer *July 2008 – March 2011*
Malware Analyst
Built machine learning systems for automated malware detection: a decision-tree system for in-lab software classification and clustering, antivirus signature technology based on locality-sensitive hashing, and an early false-positive detection system, deployed on millions of devices.

SELECTED PATENTS

[P1] M. Zaiss, F. Glang, **Sergey Prokudin**, and K. Scheffler. “Machine learning based processing of magnetic resonance data, including uncertainty quantification.” *U.S. Patent No. 11,965,946*. 2024.

[P2] **Sergey Prokudin**, J. Romero Gonzalez-Nicolas, and Michael J. Black. “Image generation from 3D model using neural network.” *U.S. Patent No. 11,403,800*. 2022.

[P3] J. Romero Gonzalez-Nicolas, **Sergey Prokudin**, and Christoph Lassner. “Rapid point cloud alignment and classification with basis set learning.” *U.S. Patent No. 11,176,693*. 2021.

[P4] **Sergey Prokudin**, and Alexey Romanenko. “System and method for distributing antivirus records to user devices.” *U.S. Patent No. 9,578,065*. 2017.

[P5] Alexey Romanenko, Ilya Tolstikhin, and **Sergey Prokudin**. “System and method for evaluating malware detection rules.” *U.S. Patent No. 9,171,155*. 2015.

PROFESSIONAL SERVICE

Area Chair: CVPR 2025.

Reviewer: CVPR (Outstanding Reviewer Award, 2023), ECCV, ICCV, ICLR, ICML, AISTATS, TPAMI, 3DV.

Workshop Organiser: “Uncertainty and Robustness in Deep Visual Learning” (CVPR 2019).

HONOURS & AWARDS

ETH Postdoctoral Fellowship	<i>2021–2022</i>
Microsoft Research PhD Scholarship	<i>2016</i>
Winner, Moscow State University Mathematics Olympiad	<i>2005</i>

PROGRAMMING SKILLS

Current: Python, PyTorch

Earlier: TensorFlow, C#, Microsoft SQL, x86 Assembler

LANGUAGE SKILLS

English (fluent), German (B1), Russian (native)